Model services — setup manual (all 3 at once)

This folder (`packages/`) is **fully self-contained and portable** — copy the *entire* `packages/` folder (all 3 service subfolders plus this manual, `requirements.txt`, `run_all.py`, `start_all.bat`/`start_all.sh`) to any computer with Python installed, and it will work with no dependency on anything else from the rest of this repository. Every model's trained weights already live inside its own `models/` subfolder — nothing outside `packages/` is needed.

This guide assumes you are **not** using Claude Code or any AI assistant — every step is a plain command you type yourself. It gets all 3 services running with **one** virtual environment and **one** command, instead of repeating setup 3 times.

What's in this folder

Item	What it is
`flood-risk-classifier/`	Flood risk (low/moderate/high) at 24h/48h/72h — its own README has full API docs
`discharge-forecaster/`	Predicted river discharge (m^3/s) at 24h/48h/72h — its own README has full API docs
`flood-susceptibility/`	Terrain-based flood-proneness score — its own README has full API docs
`requirements.txt`	One consolidated dependency list covering all 3 services
`run_all.py`	Starts all 3 services at once, in one terminal, with labeled output
`start_all.bat` / `start_all.sh`	Double-click/no-typing shortcuts for `run_all.py` (Windows / Mac-Linux)

Each service subfolder is still independently runnable on its own too (see that subfolder's own README) — this manual is just the fast path for running all three together.

Step 1: Check you have Python

Open a terminal (Command Prompt, PowerShell, or a Mac/Linux terminal) and type:

```
python --version
```

You need Python 3.10 or newer. If that command isn't found, try `python3 --version` instead. If neither works, install Python from python.org first (on Windows, check the box that says "Add Python to PATH" during installation).

Step 2: Open a terminal inside this folder

Navigate to wherever you copied the `packages` folder, e.g.:

```
cd path/to/packages
```

Everything below assumes your terminal is sitting inside this exact folder.

Step 3: Create ONE shared virtual environment

```
python -m venv .venv
```

This creates a `.venv` folder inside `packages/`, shared by all 3 services. You only need to do this once.

Step 4: Activate it

Windows (Command Prompt):

```
.venv\Scripts\activate.bat
```

Windows (PowerShell):

```
.venv\Scripts\Activate.ps1
```

Mac/Linux:

```
source .venv/bin/activate
```

You'll know it worked because your terminal prompt now starts with `(.venv)`. Do this every time you open a new terminal to work with these services — it doesn't stay active permanently.

Step 5: Install everything, once

```
pip install -r requirements.txt
```

This single install covers all 3 services (takes a minute or two, needs an internet connection).

Step 6: Start all 3 services

```
python run_all.py
```

Or, without typing a command: double-click `start_all.bat` (Windows) or run `./start_all.sh` (Mac/Linux) — both just call `run_all.py` using the venv you just set up.

You'll see labeled output like:

```
[classifier] starting on port 8000 ...
[discharge] starting on port 8001 ...
[susceptibility] starting on port 8002 ...

All 3 services starting. Once ready:
  classifier      -> http://127.0.0.1:8000/docs
  discharge       -> http://127.0.0.1:8001/docs
  susceptibility  -> http://127.0.0.1:8002/docs
```

Leave this terminal window open — all 3 keep running as long as this command is active. Press `Ctrl+C` once to stop all three together (not three separate Ctrl+C's).

Step 7: Check it's actually working

Open a **new** terminal window (leave `run_all.py` running in the first one) and either:

- Open your browser to any of the 3 `/docs` URLs printed above — each gives an interactive page to try that service's API directly, no coding needed.
- Or run these: `curl "http://127.0.0.1:8000/predict?station_id=SW90" curl "http://127.0.0.1:8001/predict?station_id=SW90" curl "http://127.0.0.1:8002/predict?station_id=SW90"`

If each returns a block of JSON (not a connection error), all 3 are working.

Pointing a dashboard at these

If you're running `frontend-glass/` or `frontend/` from this same repo, it already expects exactly these 3 ports (`8000`/`8001`/`8002`) — no configuration needed, just have `run_all.py` running before you open the dashboard and click any of the model buttons.

Troubleshooting

- **`ModuleNotFoundError`** — you forgot to activate the virtual environment (Step 4) before running `run_all.py`, or forgot Step 5's `pip install`.
- **"expected folder not found"** from `run_all.py`** — you're running the command from somewhere other than inside the `packages` folder itself. `cd` into it first (Step 2).
- **One service's `[prefix] exited unexpectedly`** — `run_all.py` deliberately stops the other two when this happens (rather than leaving a half-working set running silently). Scroll up in the same terminal for that service's own error output just above the "exited unexpectedly" line — it's the same error you'd see running that one service standalone (see its own README's Troubleshooting section).
- **A `models/` folder is missing inside one service** — make sure you copied the *entire* `packages` folder, including every subfolder's `models/` directory, not just the `.py` files.
- **Predictions take a few seconds** — normal; each call fetches fresh live weather/river data from a free public API before predicting.
- **`502` errors from a service** — the free weather API it depends on (Open-Meteo) is temporarily unreachable. Wait a minute and try again.

For anything specific to one service's API (request/response shape, example calls from JavaScript, deployment notes), see that service's own `README.md` — this manual only covers getting all three running together.